noc25_ee134

Mock Exam 5 | Course ID: noc25_ee134

Q1. What does EVSE stand for?

A) Electric Vehicle Safety Equipment
B) Electric Vehicle Supply Equipment
C) Electronic Voltage Source Emitter
D) Energy Vehicle Service Enclosure

Source: -GZyYmm0B2M | Difficulty: Easy

Q2. What is Zero Voltage Switching (ZVS)?

A) A method where the current through the switch is zero before it turns on.
B) A method where the voltage across the switch is driven to zero before the current begins to rise during turn-on.
C) A technique to eliminate all voltage from the converter circuit.
D) A failure mode where the switch voltage drops to zero unexpectedly.

Source: -kxjMKhO51M | Difficulty: Easy

Q3. What is the primary function of a capacitive filter in a rectifier circuit?

A) To increase the peak input current.
B) To store energy and release it during voltage dips, smoothing the DC output.
C) To block all DC components from the output.
D) To invert the polarity of the AC input.

Source: 1Ae4n3hWxwY | Difficulty: Easy

Q4. What does OCPP stand for?

A) Open Circuit Power Protocol
B) Optimal Charging Point Protocol
C) Open Charge Point Protocol
D) Official Charger Port Protocol

Source: 4TnD50QrkVg | Difficulty: Easy

Q5. In AC Type-2 charging, what method is used for communication on the Control Pilot (CP) line?

A) CAN bus communication
B) Power Line Communication (PLC)
C) Pulse Width Modulation (PWM) signaling
D) Ethernet communication

Source: 6-xQQwjzPIc | Difficulty: Easy

Q6. What is the primary purpose of introducing an air gap into the core of a filter inductor?

A) To increase the core's permeability.
B) To make the inductor lighter.
C) To prevent core saturation and increase the saturation current.
D) To maximize core losses for thermal management.

Source: 6X9uRXj0MZQ | Difficulty: Easy

Q7. What is the primary advantage of a Totem Pole PFC converter over a traditional boost PFC?

A) It can only operate in Discontinuous Conduction Mode.
B) It eliminates the input diode bridge, reducing conduction losses and improving efficiency.
C) It requires a much larger inductor for the same power level.
D) It operates at a lower output voltage.

Source: RxJCQteGJQA | Difficulty: Easy

Q8. In an AC charging system, where does the conversion from AC to DC power take place?

A) Inside the AC EVSE (charging station).
B) Within the charging cable itself.
C) At the grid substation.
D) Inside the vehicle, within the Onboard Charger (OBC).

Source: dHc3-Osl9E0 | Difficulty: Easy

Q9. What is the purpose of "dead time" in a half-bridge converter leg?

A) To increase switching losses for better thermal performance.
B) A period where the converter is completely powered down for safety.
C) An intentional delay to prevent shoot-through and enable soft switching.
D) The time it takes for the inductor current to reach zero.

Source: gLUPdxAcUN8 | Difficulty: Easy

Q10. What communication protocol is exclusively used for data exchange between the EV and EVSE in the BEVC-DC001 standard?

A) Power Line Communication (PLC)
B) Ethernet
C) Controlled Area Network (CAN)
D) Pulse Width Modulation (PWM)

Source: kd8MFLt2h9k | Difficulty: Easy

Q11. What is the primary difference between AC and DC charging in terms of power conversion location?

A) AC charging converts power at the grid, while DC charging converts it in the vehicle.
B) AC charging converts AC to DC inside the vehicle (onboard), while DC charging converts it outside the vehicle (off-board).
C) Both AC and DC charging convert power inside the vehicle, but at different speeds.
D) DC charging uses AC power directly, while AC charging requires an initial DC conversion.

Source: -GZyYmm0B2M | Difficulty: Medium

Q12. Why is the simplified power balance approach inadequate for sizing the DC-link capacitor in a three-phase AC-DC converter?

A) It overestimates the required capacitance by a factor of three.
B) It only works for single-phase converters.
C) It neglects the impact of high-frequency switching harmonics which cause ripple currents in the capacitor.
D) It assumes the input power is always constant, which is incorrect.

Source: -dgKwI8Qoqs | Difficulty: Medium

Q13. What is the main difference in the primary voltage applied to the transformer between a full-bridge and a half-bridge converter?

A) The full-bridge applies the full input voltage (Vin), while the half-bridge applies half the input voltage (Vin/2).
B) The full-bridge applies a sinusoidal voltage, while the half-bridge applies a square wave voltage.
C) The full-bridge applies Vin/2, while the half-bridge applies the full Vin.
D) There is no difference; both apply the full input voltage Vin.

Source: 38D14JN-3wg | Difficulty: Medium

Q14. What is the significance of the cross-coupling terms (e.g., ω * L * Isq) in the DQ axis model of a three-phase converter?

A) They represent the power losses in the converter.
B) They indicate that the d-axis and q-axis are inherently independent.
C) They link the d-axis and q-axis dynamics, and must be compensated for to achieve true independent control.
D) They are artifacts of the transformation and can be ignored in the control design.
Source: 3TF3ZUAokeo | Difficulty: Medium

Q15. What is the impact of the "D-loss" term on the effective duty cycle of a PSFB converter?

A) It increases the effective duty cycle, leading to a higher output voltage.
B) It has no impact on the effective duty cycle, only on the losses.
C) It reduces the effective duty cycle, and this reduction varies with inductor current and input voltage.
D) It is a constant loss term that only affects efficiency.
Source: 568-ZYztUIY | Difficulty: Medium

Q16. How does the EV signal its readiness for charging to the EVSE in an AC Type-2 system?

A) By sending a high-frequency signal over the power lines.
B) By changing the PWM duty ratio on the CP line.
C) By changing the voltage level of the PWM signal on the CP line (e.g., from +9V to +6V).
D) By changing the resistance on the PP pin.
Source: 6-xQQwjzPlc | Difficulty: Medium

Q17. Why are core losses often considered negligible in the design of filter inductors for DC-DC converters?

A) Filter inductors use cores that have no magnetic properties.
B) The large DC bias current completely eliminates the BH loop.
C) The AC ripple current is typically very small, resulting in a very small BH loop area and thus minimal core loss.
D) The air gap in the core redirects all magnetic flux, preventing any core losses.
Source: 6X9uRXj0MZQ | Difficulty: Medium

Q18. What is the purpose of the saturation block in the voltage control loop of a boost PFC converter?

A) To saturate the inductor core for faster response.
B) To limit the output voltage to a safe level.
C) To limit the output of the voltage controller, thereby limiting the maximum current demand to protect components.
D) To ensure the voltage controller's output is always a sinusoidal waveform.
Source: BTGv9f8yZDI | Difficulty: Medium

Q19. What is the main advantage of operating a boost PFC converter in DCM regarding the control strategy?

A) It allows for higher power transfer than CCM.
B) It completely eliminates Total Harmonic Distortion (THD).
C) It simplifies control by eliminating the need for a multiplier block and unit sine wave generation.
D) It requires a much smaller output capacitor.
Source: FBut4yhA4q0 | Difficulty: Medium

Q20. What is the key advantage of using JSON over WebSocket compared to SOAP for OCPP communication in terms of scalability?

A) JSON messages are encrypted by default, while SOAP messages are not.
B) A persistent WebSocket connection allows multiple EVSEs to be behind a single router without needing unique IP addresses for each.
C) SOAP can only handle a single EVSE per CMS.
D) JSON uses less electricity for data transmission.

Source: WUYG4Eko3rU | Difficulty: Medium

Q21. In a DC charging scenario, what is the role of the vehicle's Battery Management System (BMS)?

A) It acts as a slave, only receiving commands from the DC EVSE.
B) It is bypassed and has no role in DC charging.
C) It acts as the master, monitoring battery parameters and dictating the required charging profile to the DC EVSE.
D) Its only role is to control the vehicle's onboard charger.

Source: a-fvKWBRT_g | Difficulty: Medium

Q22. What is the purpose of the pre-charging stage in the DC CCS2 charging sequence?

A) To test the maximum current the cable can handle.
B) To gradually raise the EVSE output voltage to match the battery voltage, preventing large inrush currents.
C) To fully charge the EVSE's internal capacitors before power transfer.
D) To allow the user to select the desired charging speed.

Source: cEaVn-3aSNk | Difficulty: Medium

Q23. According to the lecture, the RMS current through the DC-link capacitor (I_C_RMS) is calculated using the formula `√(I_converter_RMS_Line^2 - I_converter_avg_Line^2)`. What does this relationship signify?

A) The capacitor current is the sum of the average and RMS converter currents.
B) The RMS capacitor current is the portion of the total RMS converter current that does not flow to the load, representing the ripple component.
C) The capacitor current is always equal to the average load current.
D) This formula calculates the peak capacitor current, not the RMS value.

Source: -dgKwI8Qoqs | Difficulty: Hard

Q24. In the iterative transformer design procedure, what is the immediate next step if the calculated total loss (`P_total`) exceeds the allowed total loss (`P_total_allowed`)?

A) Increase the switching frequency to reduce core size.
B) Select a smaller wire gauge to fit more turns in the window area.
C) Go back to Step 1 and select the next bigger core size.
D) Increase the peak AC flux density (deltaB) to reduce copper losses.

Source: 24SWNw-ePds | Difficulty: Hard

Q25. Why does the output inductor and capacitor in a full-bridge converter experience a ripple frequency of 2fsw?

A) Because the transformer has two windings, doubling the frequency.
B) Because the full-bridge rectifier on the secondary side inverts the negative half-cycles, causing the inductor current to repeat its pattern every half switching cycle (Ts/2).
C) It is an error in the analysis; the ripple frequency is always fsw.
D) Because there are four switches, and each one contributes to the ripple.

Source: 38D14JN-3wg | Difficulty: Hard

Q26. In the small signal model for a PSFB converter, how are the `DI` and `DV` terms, which account for D-loss variation, incorporated into the equivalent circuit?

A) They are ignored as they are second-order effects.

B) They are added as additional series resistors in the inductor path.
C) They are represented by additional controlled voltage and current sources that modify the basic buck converter model.
D) They only affect the steady-state DC operating point, not the small signal model.
Source: 568-ZYztUIY | Difficulty: Hard

Q27. Why does DCM operation in a boost PFC converter typically result in a higher Total Harmonic Distortion (THD) compared to CCM operation?

A) Because the switching frequency in DCM is much lower.
B) Because the average input current, while following the voltage shape, is not perfectly sinusoidal due to the triangular nature of the inductor current pulses.
C) Because DCM operation requires a larger inductor, which introduces more distortion.
D) Because the output voltage ripple is higher in DCM, which reflects back to the input current.
Source: FBut4yhA4q0 | Difficulty: Hard

Q28. In Type-2 PWM for a full-bridge converter, why is ZVS turn-on reliably achieved for the leading leg switches but challenging for the lagging leg switches?

A) The leading leg switches operate at a lower frequency.
B) The lagging leg switches have much larger output capacitances.
C) The leading leg uses energy from the large output inductor for ZVS, while the lagging leg only has the smaller leakage inductance energy available.
D) The leading leg switches are made of a different material (e.g., GaN) that switches faster.
Source: XQsOeitdyiE | Difficulty: Hard

Q29. In Single Phase Shift (SPS) modulation for a DAB converter, why is it crucial to maintain a symmetrical inductor current waveform?

A) To ensure the output voltage is always positive.
B) To maximize the power transfer at all times.
C) To prevent a DC offset in the transformer primary current, which could lead to flux walking and core saturation.
D) To simplify the calculation of RMS current.
Source: XsicKDSRQhk | Difficulty: Hard

Q30. In a balanced three-phase AC-DC converter operating at unity power factor, why is the instantaneous input power constant?

A) Because the three-phase voltages and currents are always constant.
B) Because the sum of the three phase powers, each containing a $\cos(2\omega t)$ term phase-shifted by 120°, results in the cancellation of all time-varying components.
C) Because the output capacitor perfectly smooths the input power.
D) This is an incorrect assumption; the input power always pulsates at twice the line frequency.
Source: inlEtBllhIE | Difficulty: Hard

Answer Key

1: B	2: B	3: B	4: C	5: C
6: C	7: B	8: D	9: C	10: C
11: B	12: C	13: A	14: C	15: C
16: C	17: C	18: C	19: C	20: B
21: C	22: B	23: B	24: C	25: B
26: C	27: B	28: C	29: C	30: B